SHARIF UNIVERSITY OF TECHNOLOGY

Department of Electrical Engineering

ELECTRONICS II

PROJECT 1

Design and Characterization of an LM741-Type Input Stage

Transistor-level bipolar amplifier: operating point, frequency response, loading, transient behavior and offset

Circuit Design and LTspice Simulation

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1. Design problem and scope

The design objective was to reproduce and study the **input-stage architecture** of a conventional LM741-type operational amplifier at transistor level. Unlike a complete commercial LM741, this circuit does not include all internal gain, compensation and high-current output stages. Accordingly, its open-loop gain, output swing and slew-rate behavior must not automatically be identified with the specifications of the complete IC. The circuit was designed for a high-impedance load and then investigated under stronger output loading.

The complete LM741 provides inverting and non-inverting inputs at pins 2 and 3, output at pin 6, negative and positive supply pins 4 and 7, and offset-null pins 1 and 5; pin 8 is not connected. The present circuit implements the input-stage function only. Standard complete-op-amp applications include summing and difference amplifiers, logarithmic amplifiers, Schmitt triggers and square-wave/integrator circuits; these applications motivate the gain, input-impedance and offset requirements but are not built as separate circuits in this project.

Table 1. Design targets and evaluation conditions.

Specified quantity	Requirement / test condition
Supply rails	+15 V and -15 V
Nominal load	1 MΩ
Open-loop voltage gain	50 dB target
Quiescent power	No more than 12 mW
Input resistance	Greater than 2 MΩ
Load sensitivity	Repeat AC measurements with 250 kΩ load
Analysis categories	DC operating point, AC frequency response, input/output impedance, transient response, output swing and harmonic content
Additional investigations	Output-buffer remedy; slew rate of a complete LM741 model; DC offset trimming

2. Circuit architecture and device models

The topology has three interacting subsystems. **Q1-Q2** form the signal-input differential pair; **Q3-Q4** are complementary active-load devices that develop the single-ended voltage response; and the remaining PNP/NPN mirror and reference branches establish the operating currents. The amplifier output is taken before a dedicated high-current output stage, which explains its sensitivity to the external resistance. Each supply rail is fixed at 15 V in magnitude.

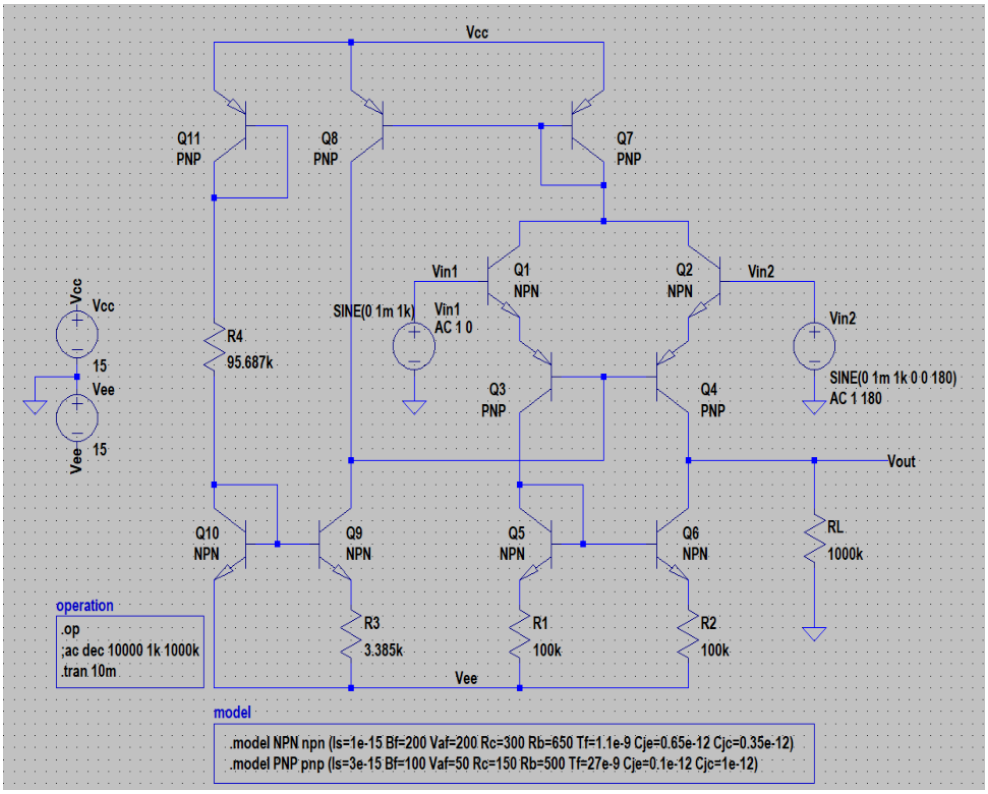


Figure 1. LTspice schematic of the transistor-level input-stage amplifier. The circuit includes both input sources, the bias network, the differential path and the external load.

2.1 Transistor and resistor values

The LTspice schematic defines custom bipolar models rather than relying on unspecified library defaults. These values were part of the project setup and affect both DC bias and frequency response. Table 2 reproduces the principal model parameters and design elements.

Table 2. Device models specified in baseCircuit.asc.

Parameter	NPN model	PNP model
Saturation current I_S	1×10^{-15} A	3×10^{-15} A
Forward gain β_F	200	100
Early voltage V_A	200 V	50 V
Base / collector resistance	650 Ω / 300 Ω	500 Ω / 150 Ω
Forward transit time τ_F	1.1 ns	27 ns
Zero-bias junction capacitances	C_{je}=0.65 pF; C_{jc}=0.35 pF	C_{je}=0.1 pF; C_{jc}=1 pF

The base schematic includes R1=R2=100 kΩ, R3=1.9555 kΩ and R4=85.1 kΩ. The saved schematic shows R_L=1 MΩ. Both input sources are sinusoidal at 1 kHz with 1 mV amplitude and opposite phase; the same pair is configured for an AC magnitude of unity each. Consequently the differential AC drive is 2 V in the frequency-domain analysis, not 1 V.

3. Bias design and small-signal analysis

For a forward-active BJT at approximately room temperature, the conventional first-order relations are

$$g_m \approx I_C / V_T \; ; \; r_\pi \approx \beta / g_m \; ; \; r_o \approx |V_A| / |I_C| \; , \; V_T \approx 25.9 \text{ mV} \; .$$

These equations provide an initial hand-design estimate. In LTspice, the actual parameters also depend on region of operation, series resistances and the full nonlinear model. The corresponding quantities were therefore taken from the LTspice operating-point log instead of being reconstructed from idealized active-region assumptions.

Table 3. Operating-point values transcribed from baseCircuit.log; current magnitudes are used for easier comparison.

Device	I_C (µA, magnitude)	V_{CE} (V)	g_m (mS)	r_π (kΩ)
Q1	13.0	+15.0	0.504	425
Q2	11.1	+15.0	0.431	497
Q3	13.0	−12.5	0.502	247
Q4	10.2	−0.070	0.428	215
Q5	12.9	+0.602	0.497	402
Q6	12.9	+13.0	0.498	426
Q7	23.7	−0.590	0.916	109
Q8	31.1	−16.2	1.20	109
Q9	32.2	+13.8	1.24	171
Q10	335	+0.687	12.9	15.4
Q11	333	−0.660	12.9	7.75

Operating-region observation. Q4 has approximately −70 mV collector-emitter voltage and an measured DC current gain of only 10.4, far below its nominal model β. This is a substantial warning that the PNP load device is not behaving as an ordinary forward-active gain element in that run. The effect cannot be neglected when interpreting gain, headroom or analytical small-signal estimates.

3.1 Supply-power budget

At zero-time operating point, the recorded supply currents are I(VCC)=−0.392 mA and I(VEE)=−0.395 mA. For two 15 V rails, the source power supplied to the circuit is approximately

$$P_Q = 15|I(VCC)| + 15|I(VEE)| \approx 11.80 \text{ mW} \; .$$

This value is below the 12 mW design limit for **the measured operating point** It is not evidence that the gain or the input-resistance targets are also satisfied.

4. Frequency response and loading

4.1 Definition of differential gain

With opposing input excitations, the correct differential voltage transfer is $H_d(j\omega)=V_{out}(j\omega) / [V_{in1}(j\omega)-V_{in2}(j\omega)]$. Reporting V_{out}/V_{in1} instead increases the apparent gain by 6.02 dB and does not characterize the intended two-input differential experiment.

$$A_{d,dB}(f) = 20 \log_{10} | V_{out}(f) / (V_{in1}(f) - V_{in2}(f)) | .$$

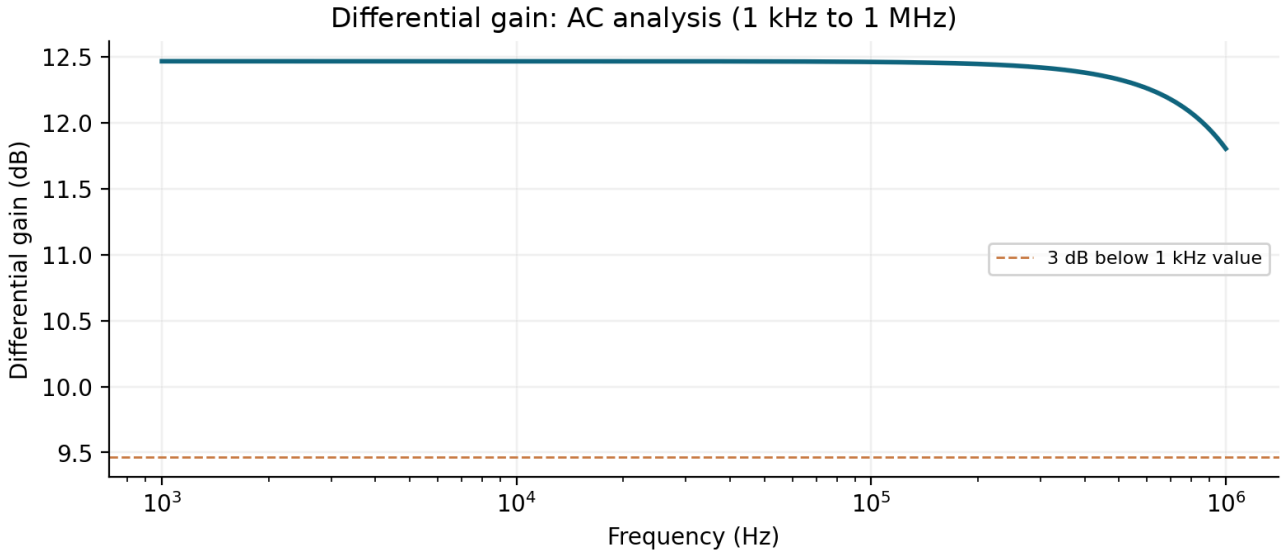


Figure 2. Differential gain recalculated from the baseCircuit.raw AC data; the stored sweep covers 1 kHz to 1 MHz.

At 1 kHz the directly reconstructed differential gain is **12.47 dB**, decreasing to **11.80 dB** at 1 MHz. No 3 dB attenuation from the 1 kHz baseline or unity-gain crossing occurs within the recorded sweep. A numeric 3 dB bandwidth or unity-gain frequency therefore cannot be established from this file alone.

4.2 Verification of the load condition

A reproducibility discrepancy is present: the stored baseCircuit.asc specifies R_L=1 MΩ, whereas the operating-point file gives V_{out}=−0.6664 V and I(RL)=−2.665 µA. Their ratio yields **approximately 250 kΩ**. This demonstrates that the stored waveform/operating-point files correspond to a more heavily loaded run than the currently saved schematic configuration. The 12.47 dB curve should therefore be interpreted as the recorded **250 kΩ-loaded case**, not as an independently verified 1 MΩ result.

Table 4. Frequency-response evidence and configuration differences.

Evidence source	Numerical observation	Interpretation
baseCircuit.asc	R_L=1 MΩ; AC 1 kHz–1 MHz	Latest saved circuit settings
baseCircuit.op.raw	$\|V_{out}/I(RL)\|$≈250 kΩ	Operating-point data from different load setting
baseCircuit.raw	A_d(1 kHz)=12.47 dB	Directly recalculated loaded transfer
Initial project report	Transient gain ≈46.46 dB; f_{3dB}≈8.45 MHz; unity crossing ≈10.46 MHz	Reported plots; the last two frequencies exceed the saved AC sweep and cannot be rederived here

4.3 Input and output impedance: measurement definitions

The intended differential input resistance is obtained by applying a small test voltage between the two input terminals, calculating the corresponding differential port current, and forming $R_{in,d}(f)=|V_d/I_d|$. Because both input sources were independently driven in the stored simulation, simply dividing a single source voltage by one source current is not a complete two-port measurement unless the chosen common-mode and differential excitation convention is explicitly retained. Output impedance requires an independent AC test current with signal inputs suppressed while holding the relevant feedback connection fixed. The initial project report contains input/output impedance screenshots, but the saved files do not preserve matching test-source configurations; the >2 MΩ constraint is therefore not presented as numerically verified.

4.4 Heavy loading and the buffer remedy

A 250 kΩ output resistor draws four times as much current as a 1 MΩ load at the same voltage. Because the examined node is the output of a high-gain, comparatively weak transistor stage, the reduced load increases current demand and degrades the attainable voltage gain. An emitter-follower buffer between the gain node and load is the natural solution: its high input impedance limits loading of the voltage-gain stage and its lower output impedance supports the external resistance.

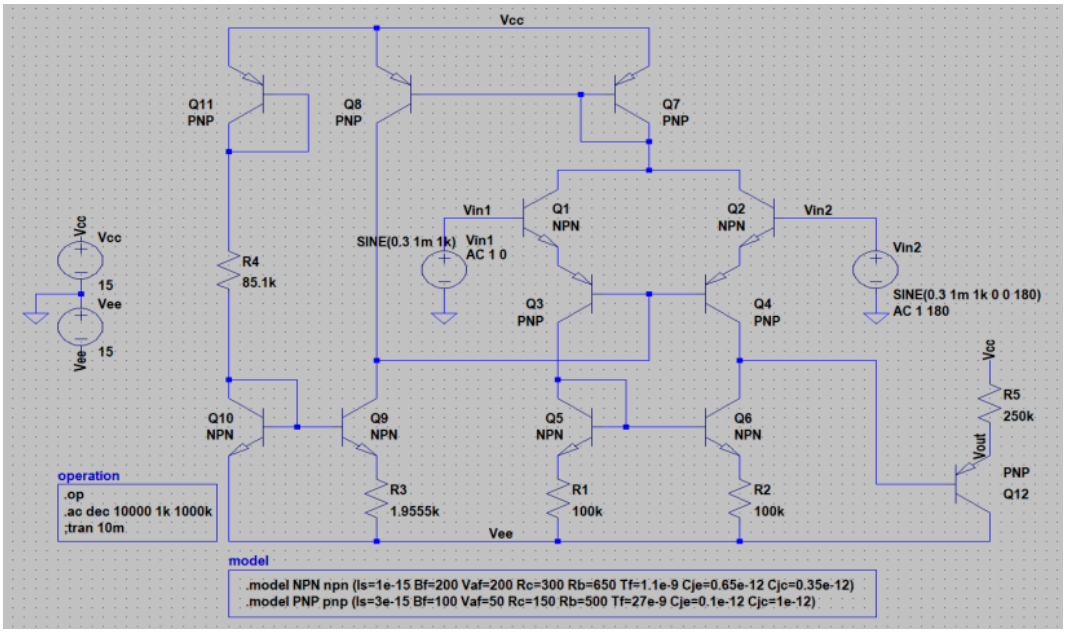


Figure 3. Buffer-assisted circuit shown in the original simulation report used to address the 250 kΩ load; The circuit configuration is shown for comparison.

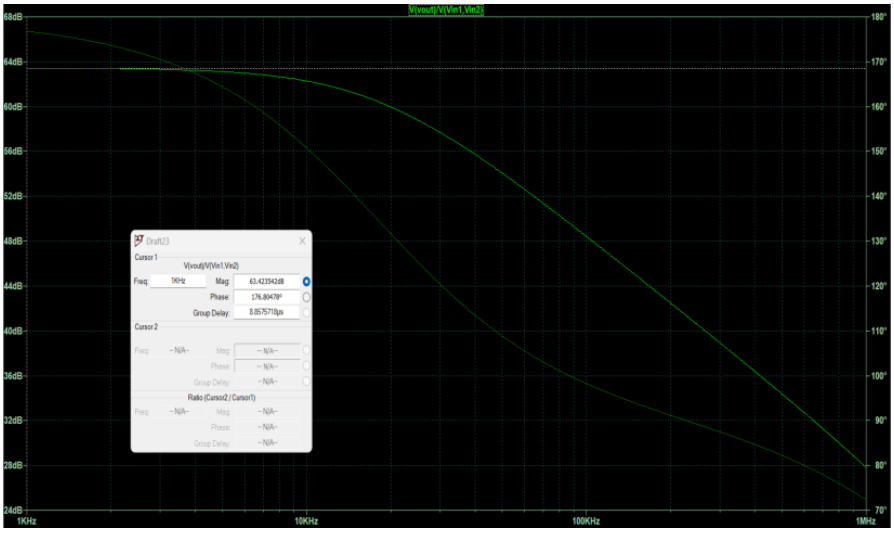


Figure 4. Gain response shown in the initial project report after adding the buffer. The exact simulation configuration for this earlier plot is not available as a separate saved .asc file.

A strict comparison of 1 MΩ, 250 kΩ and buffered 250 kΩ cases would require fresh runs under identical bias and source settings. The simulation plots establish that a buffer approach was examined, but do not by themselves determine a corrected gain or bandwidth with the same evidential strength as the stored AC waveform.

5. Transient behavior, swing and harmonic content

The project also investigated a 1 kHz sinusoidal differential drive in the transient domain. The initial project report gives an output amplitude of approximately 838.71 mV for a cited input amplitude of 3.986 mV, producing a reported ratio of about 210.4 (46.46 dB). This is a reported measurement from the report and should not be equated with the 12.47 dB loaded AC run, which was stored with a different configuration.

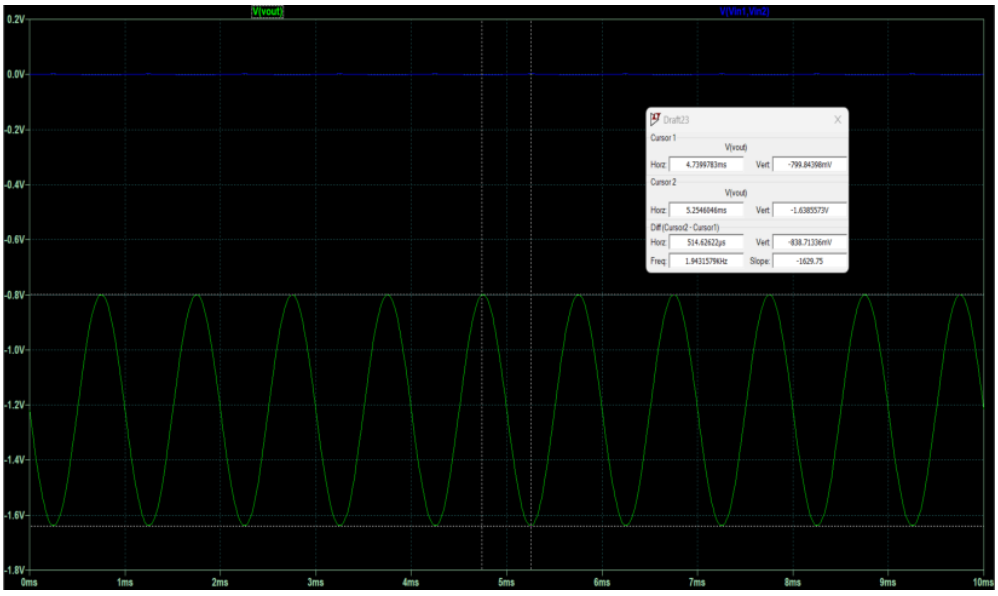


Figure 5. Report-based transient waveform used in the gain study. Cursor-based amplitudes are recorded in the report, not in a corresponding complete transient RAW for the base circuit.

For maximum linear swing, the original simulation varied the excitation from approximately 1 to 3 mV and compared the resulting waveforms. The transient analysis identifies an approximately **457 mV one-sided linear headroom** around its nonzero output bias; this is not the same as a 457

mV peak-to-peak swing. The result also depends on where clipping, asymmetry and the DC baseline are defined.

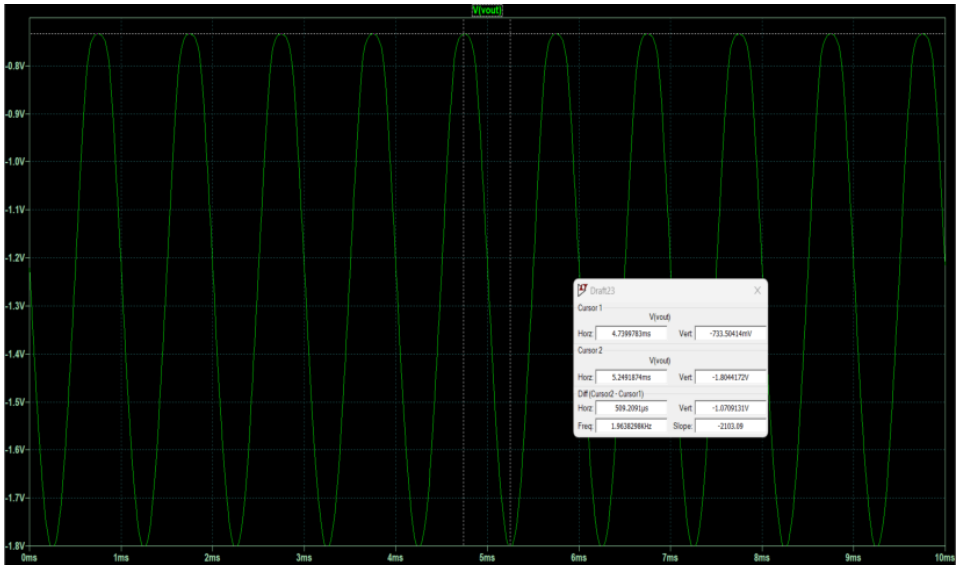


Figure 6. Transient waveform from the sweep of drive amplitudes used to evaluate the limits of linear output swing.

5.1 Harmonic interpretation

A purely linear amplifier transfers a 1 kHz sinusoid primarily at 1 kHz. As the input drives the transistors toward saturation or cutoff, harmonics at integer multiples of the input frequency become visible. The initial project report includes FFT figures at a small drive and at a larger, nonlinear drive. These plots support a qualitative distortion comparison; no reproducible numeric THD for both conditions is preserved in the project files.

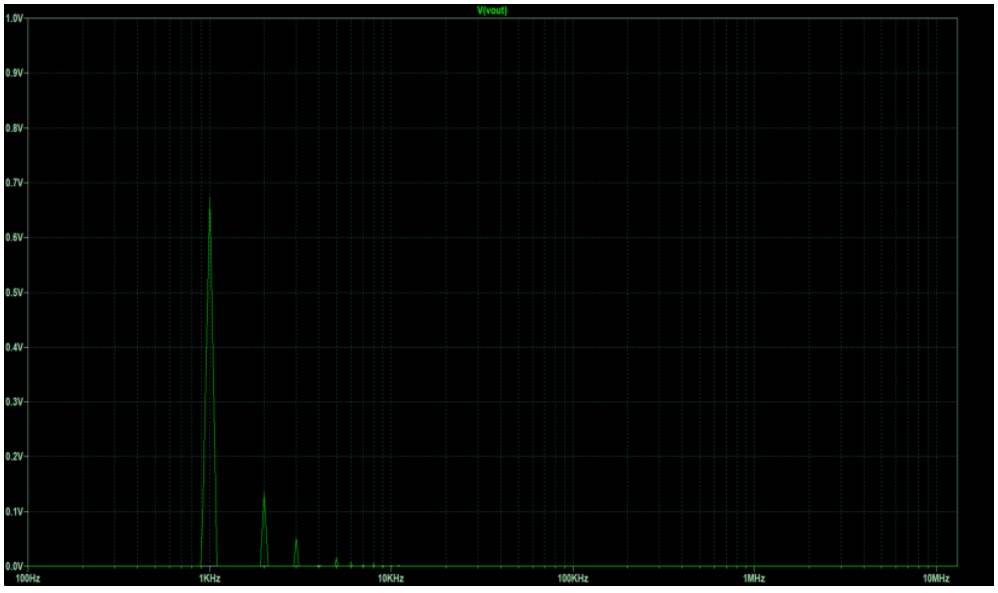


Figure 7. Output spectrum for a larger excitation. Higher harmonics arise from the nonlinear transfer near the swing limits.

6. Slew rate and DC offset

6.1 Slew-rate experiment

Slew rate is the maximum absolute slope of a large-signal output transition, $SR=\max|dV_{out}/dt|$. The slew-rate experiment is implemented with an **LM741 macromodel symbol** and a fast input pulse, not with the transistor-level circuit documented in Section 2. The initial report gives approximately 0.317 V/µs, which is consistent in order of magnitude with an LM741-type macromodel; the number must not be attributed to the isolated input-stage design.

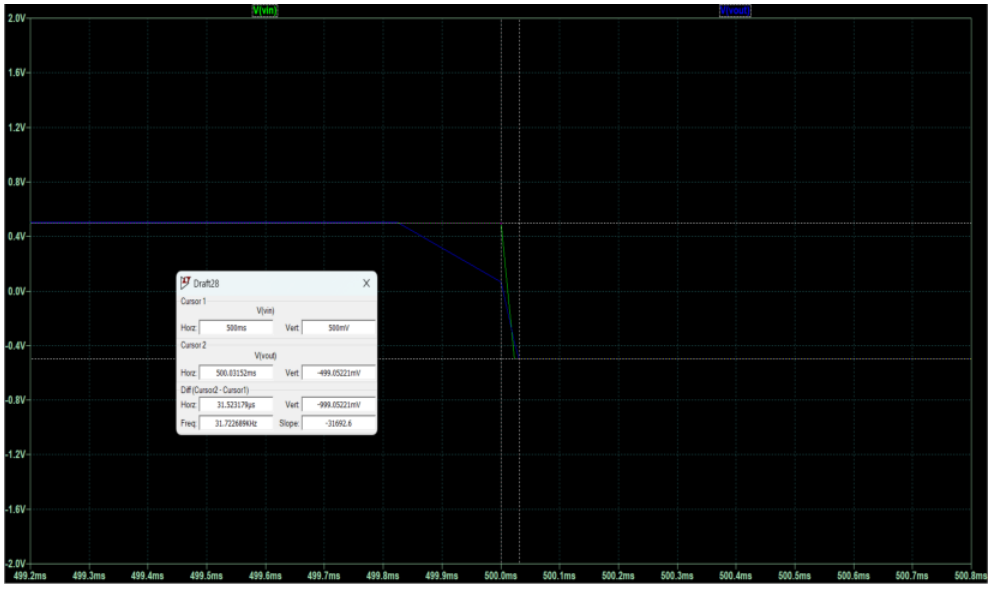


Figure 8. Report-based large-signal waveform for the standalone LM741 macromodel used in the slew-rate exercise.

6.2 Output-offset adjustment

The source design adds a **50 kΩ trim resistor R5** in the bias/output branch of the transistor-level circuit. The saved adjusted operating point is $V_{out}=-0.6032$ V; the saved short transient remains near −0.603 V. The initial project report also describes an earlier baseline near −1.2 V, but the recorded base operating-point file instead gives −0.6664 V under an inferred 250 kΩ load. Since the saved configurations differ, these values are not a controlled before/after comparison of only the trim resistor.

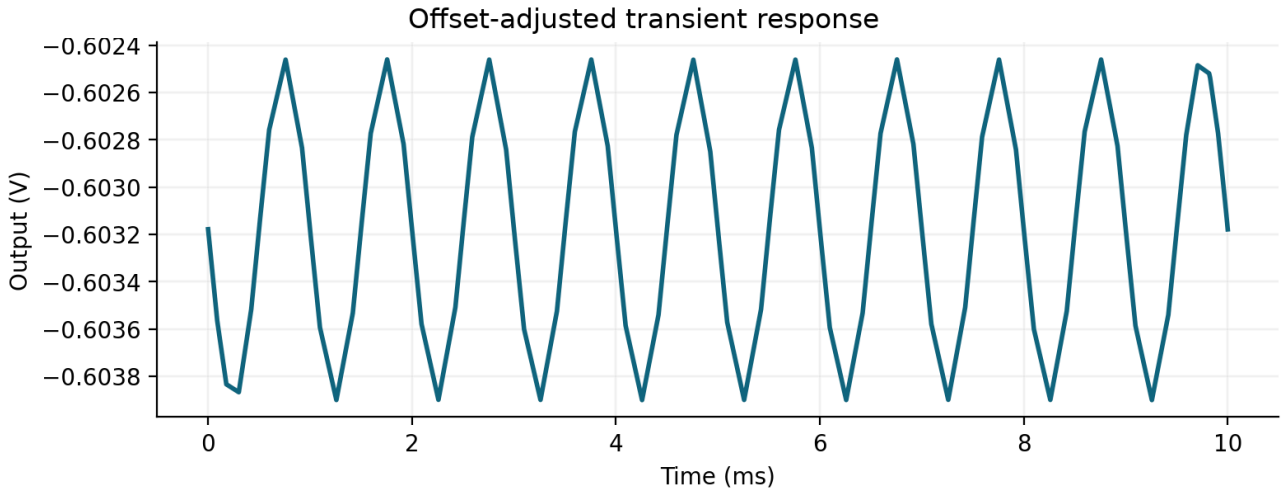


Figure 9. Output voltage reconstructed from offsetImprovingCircuit.raw. The steady level is approximately −0.603 V for the saved adjustment run.

7. Verification against design requirements

Table 5. Verification of specified design targets and measurements.

Criterion	Evidence in the available files	Assessment
Supply, nominal load and model constraints	±15 V and transistor models in the saved schematic	Established as design setup
Quiescent power ≤12 mW	11.80 mW from recorded base operating-point currents	Met for that saved 250 kΩ-loaded operating point
Gain target 50 dB	Reported transient 46.46 dB; directly recovered loaded gain 12.47 dB	Not demonstrated as met
Input resistance >2 MΩ	No matched differential input-impedance sweep with complete source configuration	Not independently verified
3 dB and unity-gain frequencies	Reported values: 8.45 and 10.46 MHz; AC file stops at 1 MHz	Not reproducible from current AC files
250 kΩ load and buffer remedy	Original diagrams and gain screenshot available	Qualitatively documented; numeric verification incomplete
Slew rate	Reported ~0.317 V/µs with complete LM741 macromodel	Model-level auxiliary experiment, not input-stage property
DC offset	Adjusted run −0.603 V, with differing base load	Adjusted output documented; controlled improvement not isolated

8. Conclusion

The saved simulation indicates 11.80 mW quiescent power, but does not establish the 50 dB gain or input-resistance targets. The 250 kΩ load discrepancy and Q4 bias condition affect interpretation of the earlier gain results. The buffer and offset-trim configurations address loading and DC offset; consistent reruns would be needed to quantify their benefits.